

COLUMBIA GORGE, WA, USA

WMO: 726988

Lat: 45.619N

Lon: 121.166W

Elev: 235

StdP: 14.57

Time Zone: -8.00 (NAP)

Period: 00-19

WBAN: 24219

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	16.8	21.4	7.8	8.3	19.9	13.3	10.9	24.0	24.6	46.1	20.1	46.7	3.1	110	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	98.2	66.6	94.3	65.8	90.4	64.8	69.4	91.0	67.8	88.7	66.3	86.4	10.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	81.7	77.6	59.6	77.0	76.4	58.1	72.7	75.3	33.5	91.4	32.1	88.7	30.9	86.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	24.0	20.9	DB	11.4	104.6	6.1	2.8	7.0	106.7	3.4	108.3	0.0	109.9	-4.4	112.0	(4)
(5)			WB	10.1	72.1	5.8	1.4	6.0	73.2	2.6	74.0	-0.7	74.8	-4.9	75.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	37.0	40.3	46.7	53.0	61.6	67.6	75.3	74.7	66.5	54.5	42.9	36.2		
	DBStd	15.10	7.10	6.69	5.95	5.50	6.44	6.39	5.58	5.32	5.81	6.20	7.50	6.91		
	HDD50	1531	405	278	131	29	1	0	0	0	0	25	232	430		
	HDD65	4616	869	693	567	361	148	41	1	1	49	329	663	894		
	CDD50	3269	1	5	28	119	360	528	785	765	494	165	18	1		
	CDD65	881	0	0	0	1	41	120	321	301	93	4	0	0		
	CDH74	10404	0	0	1	80	627	1409	3736	3341	1152	58	0	0		
	CDH80	4861	0	0	0	18	228	611	1907	1645	442	10	0	0		
	WSAvg	7.8	4.1	4.9	6.9	9.3	10.6	12.2	12.7	11.0	8.2	5.6	4.2	4.1		
	PrecAvg	14.4	2.8	1.6	1.3	0.7	0.4	0.4	0.2	0.4	0.5	1.0	2.2	3.0		
(15) Precipitation	PrecMax	19.6	7.6	4.3	3.2	1.5	1.3	1.0	0.7	1.9	1.3	2.8	5.8	9.1		
	PrecMin	7.6	0.4	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.3		
	PrecStd	3.3	1.7	1.1	0.8	0.5	0.4	0.3	0.2	0.5	0.4	0.8	1.3	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.9	71.3	82.8	92.3	99.4	103.4	102.9	94.8	80.6	64.8	55.7		
		MCWB	49.7	51.4	54.7	59.2	63.9	66.6	67.8	66.8	65.4	60.0	55.7	50.0		
	2%	DB	52.8	56.7	65.0	74.9	86.3	92.4	98.6	97.7	89.6	73.9	60.3	51.4		
		MCWB	46.9	48.7	51.3	55.7	62.0	65.4	66.9	66.0	63.1	57.6	53.1	46.3		
	5%	DB	49.3	53.6	61.1	69.4	81.0	86.9	94.9	93.5	85.4	69.4	56.9	48.3		
		MCWB	44.6	45.9	49.8	53.5	59.9	62.7	66.3	65.7	62.2	55.5	49.9	43.9		
	10%	DB	46.2	50.4	57.4	64.9	76.0	82.1	90.3	89.2	80.9	65.9	53.6	45.3		
		MCWB	42.7	43.8	48.2	51.5	57.9	61.5	65.6	64.6	60.5	54.3	48.1	41.7		
	0.4%	WB	50.9	52.6	56.5	60.3	66.2	69.9	71.5	71.2	68.0	62.7	58.3	51.3		
		MCDB	56.1	58.9	68.0	78.5	86.5	92.3	94.1	92.5	87.6	75.3	63.3	54.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.6	49.5	53.1	57.2	63.5	66.8	69.7	69.1	65.6	59.7	53.7	46.9		
		MCDB	52.0	56.1	61.7	71.0	83.5	87.8	92.4	89.5	84.2	69.8	58.4	50.5		
	5%	WB	45.4	46.6	51.0	54.6	61.2	64.6	68.0	67.5	63.8	57.6	51.1	44.3		
		MCDB	49.0	52.3	58.9	67.8	77.7	84.5	89.8	87.8	81.0	66.9	55.7	47.9		
	10%	WB	42.9	44.6	48.9	52.4	59.2	62.4	66.4	66.0	62.0	55.7	48.7	41.9		
		MCDB	46.2	49.5	56.7	63.2	73.4	79.5	87.2	85.4	78.0	64.2	52.9	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	15.5	19.2	21.2	23.3	22.9	26.2	26.2	26.1	21.3	14.3	9.8		
		MCDBR	15.9	20.6	25.9	31.5	34.0	33.3	34.6	35.0	35.2	27.7	18.2	14.7		
		MCWBR	11.0	13.0	14.7	15.3	13.9	12.1	12.0	12.0	14.1	13.9	11.4	10.3		
		MCDBR	15.3	17.9	22.5	27.4	28.7	29.4	30.1	29.2	29.6	22.6	16.3	14.2		
	5% WB	MCWBR	10.9	11.7	13.3	13.7	12.2	11.3	10.9	10.5	12.3	12.3	10.9	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.325	0.338	0.374	0.387	0.374	0.368	0.377	0.361	0.352	0.337	0.325		
	taud		2.417	2.479	2.461	2.346	2.336	2.392	2.427	2.409	2.431	2.441	2.423	2.403		
	Ebn at Noon		245	272	284	281	279	283	283	277	272	257	238	231		
	Edh at Noon		23	26	30	37	39	37	35	35	31	27	23	21		
(44) All-Sky Solar Radiation	RadAvg		390	708	1104	1618	1985	2255	2439	2058	1532	923	498	320		
	RadStd		51	98	85	115	135	106	99	108	83	83	48	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+79

Nomenclature: See separate page